

Robot actuators as a business in India

The robot market to 2030–2035, copper, software vs hardware, who makes actuators in India, how actuator factories are built, how much capital it takes, and a go-to-market plan

US\$8–14 bn

humanoid-actuator market in 2030
(from $\approx$ US\$0.23 bn in 2025);
US\$10–90 bn in 2035

< 0.1 %

humanoids' share of world copper
demand in 2030: copper is not the
lever

≈ 0

Indian makers of RobStride-class
actuators at volume; $\approx$ 55k
industrial servo joints/yr imported

₹105.9 cr

external capital from garage to a
60k/yr plant, in four stages

₹126.0 cr/yr

revenue at 60k actuators/yr ($\approx$
US\$13 M)

≈ 25 %

of the whole 2030 actuator market
needed for a US\$10 bn company at
4 $\times$ sales

Summary. Making actuators ("selling shovels") is a sounder business than making another humanoid: the market is real, growing $\approx 40\times$ from 2025 to 2030, and every Indian robot maker imports its joints today. But it is a manufacturing business with 25–45 % gross margins, against Chinese makers already at 50k–500k units a year. An Indian company should start as an integrator and supplier to labs, startups and defence, sell trust, service and customisation rather than price, use grants before venture capital, and build volume tooling only against signed orders. Copper will likely stay expensive, but because of grids, EVs and data centres, not robots.

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1. Your questions, answered straight

Question	Short answer
Is an actuator company a good business?	It is a real and growing business: every robot needs 20–40 actuators and banks put the humanoid-actuator market at $\approx$ US\$8–14 bn in 2030 (from $\approx$ US\$0.23 bn in 2025). But it is a hard, low-margin manufacturing business against very strong Chinese incumbents (RobStride, Encos, Unitree make 50k–500k joints a year at $\approx$ ¥1,000 each). It is a good business for India only if you pick buyers who will not or cannot buy Chinese, and if you scale slowly with customer money, not hype.
"1 % of a trillion dollars = US\$10 bn"?	The trillion-dollar numbers are the <i>whole robot market in 2050</i> (Morgan Stanley US\$5 trn, Citi US\$7 trn), not something you can take 1 % of today. 1 % of the 2030 humanoid-actuator market is $\approx$ US\$104 M of revenue a year, worth $\approx$ US\$416 M at a typical 4 $\times$ sales. A US\$10 bn company needs $\approx$ US\$2.5 bn of revenue: about a quarter of the entire 2030 market, or 10 million actuators a year. Possible only in a 2035+ world, and only for a top-3 global supplier.
Will copper prices go up because of robots?	Copper probably will stay strong (S&P sees a 25 % supply gap by 2040) but not because of robots . A humanoid holds $\approx$ 6–8 kg of copper ($\approx$ US\$115, < 1 % of its cost). Even bullish forecasts make humanoids 0.02–0.08 % of world copper demand in 2030 and < 0.35 % in 2035. Grids, EVs (83 kg each) and AI data centres (20–40 t per MW) drive copper. For an actuator maker copper is a small cost, not a lever.
2030: will coding stay, or will robot manufacturing win?	Both stay, but differently. AI is already shrinking <i>entry-level</i> coding jobs (US developers aged 22–25 down $\approx$ 20 % since 2024, Stanford AI Index 2026) while total software jobs still grow (BLS +15.8 % to 2034). Physical manufacturing know-how is much less exposed. The strongest position is both : a CS person who understands motors, control and factories. In actuators the differentiator is increasingly software: FOC firmware, calibration, digital-twin models and fleet data.
Should we make actuators instead of whole robots?	Yes, this is the right instinct ("sell shovels in a gold rush"): every Indian robot maker needs actuators, all of them import today, and a component supplier is not betting on which robot company wins. Start as the Indian actuator supplier + integrator , not as another humanoid startup.

2. The robot market to 2030–2035

Humanoids went from a few hundred units in 2023 to $\approx$ 13,000–20,000 in 2025 (Omdia; BofA), almost all made in China (AgiBot $\approx$ 5,200, Unitree $\approx$ 4,200–5,500, UBTEch $\approx$ 1,100). Unitree's 2025 revenue was ¥1.7 bn at a 60 % gross margin; its humanoid price fell 72 % in two years to $\approx$ US\$25k. Tesla says meaningful Optimus volume comes in 2027.

Forecast (label)	2030	2035	Long run
BofA Institute, Mar 2026 (VERIFIED)	1.2 M humanoids / yr; BOM < US\$17k	10 M / yr	3 bn in use by 2060
Goldman Sachs, Sep 2026 (press)	0.89 M units	6.5 M units, US\$138 bn	–
Morgan Stanley (press)	US\$28 bn; China 446k units	13 M in use	$\approx$ 1 bn units, US\$5 trn by 2050
UBS (press)	–	2 M in use, US\$30–50 bn	300 M by 2050
Citi GPS, Nov 2024 (VERIFIED)	–	–	648 M units, US\$7 trn by 2050

Table 1. Humanoid forecasts (research/raw/market_outlook_2030_copper_raw.md). 2035 estimates differ by $\approx$ 5 $\times$; all banks raised them in the last year.

The actuator slice. In BofA's 2030 bill of materials, rotary + linear actuators are 51 % of a humanoid (hands another 19 %). That gives an actuator market of $\approx$ **US\$0.23 bn in 2025, US\$8–14 bn in 2030**

and US\$10-90 bn in 2035 depending on the bank (ESTIMATED).

Industrial robots are the steadier market: > 600k installed worldwide in 2025 (IFR, VERIFIED), 806k forecast for 2029. China installed 354k; **India ≈ 9-10.5k (6th in the world)**, i.e. roughly 55,000 servo joints a year, **all imported**. India's domestic humanoid demand to 2030 is likely only thousands of units (Addverb targets 3,000 installs; MeitY announced a ₹500 cr humanoid fund in June 2026). An Indian actuator company therefore needs **export customers and non-humanoid volume** (AMRs, cobots, exoskeletons, defence UGVs, servo replacement) from day one.

3. Copper: a good bet, but not because of robots

Item	Value	Source / label
World refined copper demand	≈ 28 Mt (2025) → ≈ 32 Mt (2030) → 42 Mt (2040)	S&P Global, Jan 2026 (VERIFIED); 2030 interpolated
Supply gap	supply peaks ≈ 33 Mt in 2030; ≈ 10 Mt (25 %) short by 2040	S&P (VERIFIED); JPMorgan ≈ 2 Mt deficit 2030 (press)
Near term	ICSG forecasts small surpluses in 2026-27	ICSG (press)
Price	≈ US\$14,500/t (Oct 2026, +30 % y/y, near record)	COMEX (VERIFIED)
Copper per battery EV / data centre	83 kg per BEV; 20-40 t per MW	ICA/IDTechEx; JPMorgan
Copper per humanoid	≈ 6-8 kg (windings 2-7 kg, harness, PCBs, battery)	ESTIMATED; CRU 4-8 kg, MS 4.5-8.5 kg
Copper in one JXA-120	≈ 0.1 kg of winding wire (≈ ₹250 at bulk prices)	our design model
Humanoids' share of copper demand	0.02-0.08 % (2030); 0.14-0.33 % (2035); ≈ 4 % only with 1 bn robots (2040+)	ESTIMATED

Bottom line: copper is the right metal to understand (it is the winding), but owning copper is an investment decision, unrelated to whether an actuator company works. For your actuators, a 30 % rise in copper adds ≈ ₹75 to a JXA-120; a change in magnet export rules can stop production entirely. **Worry about magnets, not copper.**

4. Software vs hardware in 2030: what the evidence says

- **Entry-level coding is being squeezed.** US software developers aged 22-25: employment down ≈ 20 % since 2024, older developers +6-12 % (Stanford AI Index 2026). Brynjolfsson et al. (2026) find young workers in AI-exposed jobs 19 % below trend, through less hiring rather than layoffs.
- **Software as a whole still grows.** BLS projects software developers +15.8 % over 2024-34 but "computer programmers" -6 %: writing code is commoditising; designing systems is not.
- **Robots replace physical work slowly.** WEF 2025 expects robotics/automation to be a small net job reducer (≈ -5 M jobs globally); each extra robot per 1,000 workers lowers employment by 0.2 points (Acemoglu & Restrepo).
- **What that means for you:** an actuator is ≈ 50 % mechanics and magnetics and ≈ 50 % software (FOC, encoder calibration, thermal models, CAN protocol, test automation, simulation models for reinforcement learning). Your CS background is an advantage. The skills AI will not quickly replace are the physical ones: winding, gear quality, supplier management, factory process. Learn those; let AI help write the code.

5. Who makes actuators in India today

Short version: **nobody makes RobStride-class actuators in India at volume.** A handful of startups build quasi-direct-drive joints by hand in tens to hundreds; everyone else imports. No Indian harmonic-

reducer maker was found. Meanwhile India *does* have the building blocks at scale in neighbouring industries: drone motors (Vector: 300k/yr, own winding), EV motors (Sona, Lucas TVS, Chara), BLDC fans (Atomberg: millions/yr) and laminations (Pitti).

Company	Product	How they make it	Scale / money	Label
xTerra Robotics (Kanpur/Bengaluru)	QDD A2: 12 N·m, 550 g, ₹94,400	Parts machined by Bengaluru aerospace job shops, assembled in-house	≈ 15 people, ≈ ₹1 cr revenue, grants only	Mixed
Bhairav Robotics (Kakinada)	'Prabal' QDD, quadrupeds, UGVs	In-house	Zen Technologies bought 45 % for ≈ ₹4 cr	UNVERIFIED
Perceptyne (Hyderabad)	Dual-arm semi-humanoid, own actuators	In-house	US\$3 M seed (Endiya, Yali), 2024	UNVERIFIED
Addverb (Noida; Reliance)	Elixis-W wheeled humanoid ≈ US\$45k; 'BLDC + planetary'	Undisclosed (likely imported joints)	FY25 revenue ₹800 cr; plant ≈ ₹75 cr for 60k robots/yr	Mixed
General Autonomy (Bengaluru)	Atom 01 humanoid, Param quadruped	Imports actuators	Seed ₹32 cr at ₹280 cr valuation (pre-revenue)	UNVERIFIED
Havells CRI	Hiring to build harmonic + QDD actuators	Starting	-	UNVERIFIED
Vector Technics (Hyderabad)	Drone propulsion motors	Winds own motors, machines own parts; 300k units/yr	Zen paid ≈ ₹25 cr for 51 %	VERIFIED
Sona Comstar / Lucas TVS / Chara	EV traction and hub motors	Automated winding + stamping lines	Sona ₹99.7 cr for 200k motors/yr; Chara ₹52 cr Series A	Mixed
Atomberg / fan makers; Pitti Engineering	BLDC fan motors; laminations (Pitti 90k t/yr)	Mass stamping and winding	Atomberg 6.6 M units/yr	Mixed
Yaskawa India (Bengaluru)	Industrial servos	Only confirmed local servo production	-	UNVERIFIED

Table 2. Indian actuator and motor landscape (research/raw/india_robotics_industry_capital_raw.md). Funding figures are mostly from press summaries.

Indian manufacturing clusters you would use: Coimbatore (motors, pumps, micro winding units), Rajkot (CNC, shafts, winders), Pune/Chakan (gears, auto parts), Bengaluru Peenya (aerospace-grade CNC), Hyderabad (drone motors, laminations, defence), Chennai-Hosur (Bonfiglioli, Delta, Sona, Agile Robots' ₹300 cr plant). Contract manufacturers: Zetwerk (≈ 7,000 suppliers) for machined parts; EMS firms (Kaynes, Syrma, SFO, Dixon) for driver boards.

6. How an actuator factory is built (and the equations behind it)

6.1 Three architectures

Architecture	What you do in-house	What you buy	Volume	Capex	Who does this
A. Design + assembly house	design, winding, assembly, calibration, test, firmware	stator cores, magnets, gears, housings, bearings, PCBs (all job work)	100–3,000 / yr	₹15.7 lakh (our garage + workshop kits)	xTerra, Unitree (still mostly manual assembly)
B. Partly integrated	+ automated needle winding, magnet bonding, gear finishing, end-of-line test	laminations, magnets, castings, bearings, PCBs	3k–30k / yr	₹1–3 cr for 10k/yr (ESTIMATED)	RobStride-type start, Indian EV-motor lines

C. Volume plant	+ stamping, die-casting, gear hobbing/PM, automated lines, own driver SMT	magnets, bearings, chips	50k-1 M / yr	₹20-100+ cr; China: ₹110 M for one joint-module line (Fulin)	Encos, Quanzhibo (90 s per joint, > 85 % automated)
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6.2 The process flow of an actuator line

Incoming inspection → lamination stack (stamp or buy) → insulate → **wind** (needle winder) → terminate → hipot/resistance test → varnish + cure → rotor (bond magnets, balance) → gear stage (finish, inspect, grease) → housing (cast + machine) → **final assembly** (stator, rotor, gears, bearing, driver PCB) → flash firmware, calibrate encoders → **end-of-line test** (back-EMF, K_t , cogging, thermal, backlash) → burn-in → pack. Chinese benchmarks: 90 s per joint, > 96 % first-pass yield.

6.3 The equations you plan with

Gross margin: $GM = (P - C) / P$ Break-even volume: $N^* = F / (P - V)$

Learning curve (Wright): $C(N) = C_1 \cdot N^{\log_2 r}$ (r = cost ratio per doubling; our model ≈ 0.83 between 25 and 1,000 units/yr)

Takt time = available time / demand Line capacity = available seconds $\times$ OEE / takt

Working capital = $COGS/365 \times (\text{inventory days} + \text{receivable days} - \text{payable days})$

Capital to raise = capex + operating losses until cash-positive + working capital + ≈ 6 months of opex

Market cap = revenue $\times$ (EV/sales multiple) revenue = robots/yr $\times$ actuators per robot $\times$ your share $\times$ price

Worked example: one line at a 90 s takt, 2 shifts $\times$ 300 days and 65 % OEE makes $\approx 124,800$ actuators a year. So capacity is not the problem; **demand, cost and quality are**. Working capital is large in this business because magnets and bearings are imported with 2-4 month lead times: we assume 120 inventory days, 60 receivable days, 30 payable days.

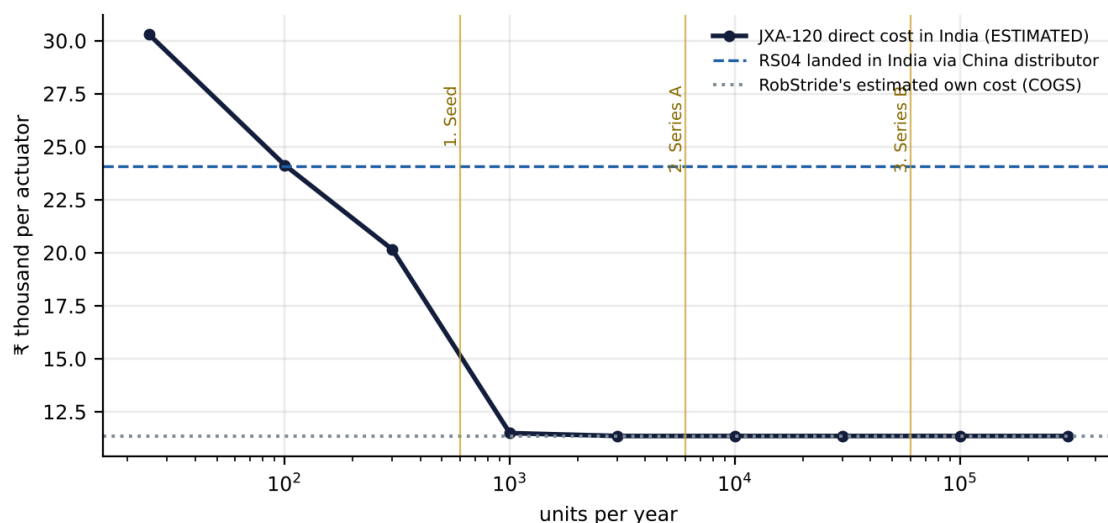


Figure 1. Our JXA-120 cost vs yearly volume (ESTIMATED): cost falls steeply until $\approx 1,000$ /yr (job shops → own tooling), then cannot fall below RobStride's own cost because magnets, bearings and chips cost everyone the same.

7. How much capital, when, and from whom

The plan below grows from a garage to a 60,000-a-year plant in four stages. It uses our own cost model and conservative assumptions (prices fall as volume rises; cost never falls below RobStride's estimated own cost). Total external capital across all stages: **≈ ₹105.9 cr** (≈ US\$11 M).

Stage	Time	Units/yr	Price	Unit cost	GM	Revenue/yr	Opex/yr	EBITDA/yr	Capex	Raise
0. Bootstrap / prototypes none yet: build JXA-120 v1/v2, test, show a walking leg	9 mo	12	-	₹30,292	-	₹0.0 cr	₹0.1 cr	₹-0.2 cr	₹0.1 cr	₹0.3 cr
1. Seed: first customers Indian university labs, robotics startups, defence R&D (iDEX), cobot/exoskeleton makers	18 mo	600	₹42,000	₹18,000	57 %	₹2.5 cr	₹1.7 cr	₹-0.2 cr	₹0.5 cr	₹2.1 cr
2. Series A: product line Indian humanoid/quadruped/AMR makers, defence programmes, export to non-China-sourcing buyers	24 mo	6,000	₹30,000	₹11,357	62 %	₹18.0 cr	₹8.6 cr	₹2.6 cr	₹4.0 cr	₹11.1 cr
3. Series B: volume plant humanoid OEMs at volume (India + export), industrial automation	24 mo	60,000	₹21,000	₹11,357	46 %	₹126.0 cr	₹38.8 cr	₹19.1 cr	₹45.0 cr	₹92.4 cr

Table 3. Staged plan (business/actuator_business_model.py; ESTIMATED). Units are JXA-120 equivalents. "Raise" = capex + losses during the stage + working capital + 6 months of opex.

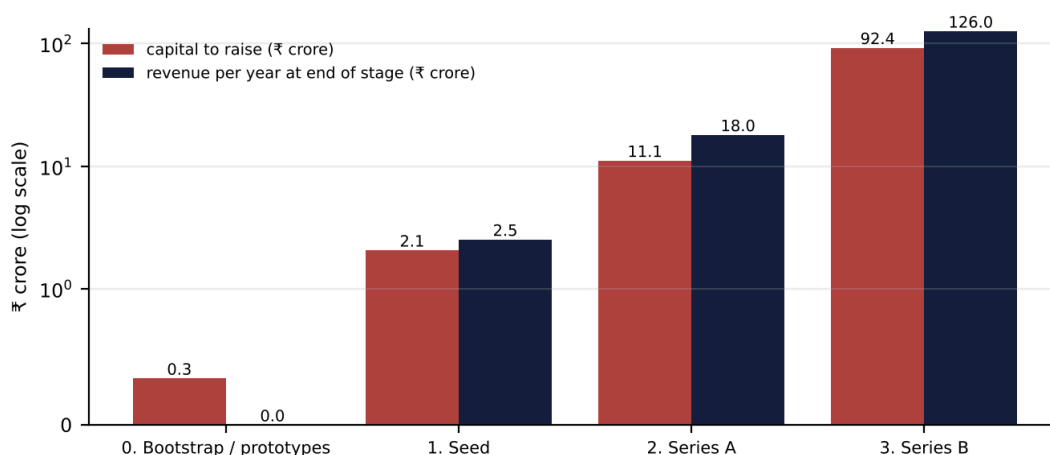


Figure 2. Capital to raise per stage vs revenue per year reached at the end of the stage (₹ crore, log scale).

- **Stage 1 is nearly break-even** only because Indian labs and defence buyers pay ≈ ₹42k for a 120 N·m joint (still less than half a CubeMars AK10-9's ₹1.05 lakh in India). It needs ≈ 692 units a year to break even.
- **The danger is Stage 3:** competing at ₹21k per unit leaves a 46 % gross margin. If Chinese makers cut prices so that you must sell at ₹15k, the margin falls to **24 %**. Do not build a volume plant without signed volume contracts.

- At the end of Stage 3 the company makes $\approx ₹126.0$ cr ($\approx$ US\$13 M) a year. At 2–4 $\times$ sales that is worth $\approx$ US\$26–53 M: a good company, not yet a unicorn.

7.1 Where the money comes from

Source of money	How much	Fit
Your savings / bootstrap	₹10–35 lakh	Stage 0: garage kit + first prototypes (this repo's numbers)
Startup India Seed Fund (via incubator)	$\leq ₹20$ lakh grant + $\leq ₹50$ lakh debt	Stage 0–1, company ≤ 2 years old
Incubators: ARTPARK (IISc), IIT hubs (DST NM-ICPS ₹3,660 cr)	grants, labs, customers	xTerra and Twara came from here
State: Karnataka ELEVATE, TANSEED	$\leq ₹50$ lakh / ₹10–15 lakh	Stage 1
Defence: iDEX / ADITI	$\leq ₹1.5$ cr / $\leq ₹25$ cr grants	strong fit: UGV, exoskeleton, robot joints with no Chinese supply
MeitY humanoid fund (announced Jun 2026)	₹500 cr total	R&D and indigenous manufacturing (UNVERIFIED details)
Seed VC: Yali, Speciale, Endiya, pi Ventures, Kalaari, Exfinity, Blume	US\$2–3.5 M typical	Stage 1
Series A/B VC: Peak XV, Elevation, Info Edge, Capital-A, TDK Ventures	US\$5–10 M / US\$20–30 M	Stage 2–3
Strategic investors: Zen (bought Vector, Bhairav), Reliance (Addverb), Zoho	acquisition or minority stake	an exit path
Debt: SIDBI venture debt, credit guarantee ($\leq ₹20$ cr), RDI scheme loans	₹10–15 cr; RDI for large projects (paused Oct 2026)	Stage 2–3 capex
ECMS component scheme (₹40,000 cr)	25 % capex support above ₹10–50 cr investment	Stage 3; ask MeitY whether robot joints qualify

Table 4. Capital sources ([india_robotics_industry_capital_raw.md](#); check official guidelines before applying). Indian robotics startups raised only US\$52.9 M in all of 2025 (≈ 0.5 % of Indian tech VC; the US raised US\$5.6 bn), so plan for **small rounds and customer revenue**.

8. Valuation reality check

Listed company	Market cap, US\$ bn	Price / sales
Harmonic Drive (JP)	3.96	10 $\times$
Nabtesco (JP)	3.7	1.9 $\times$
Leaderdrive (CN, harmonic reducers)	7.5	75 $\times$
Sanhua (CN)	21.3	4.5 $\times$
Tuopu (CN)	12.0	2.6 $\times$
Inovance (CN, servos)	20.9	2.9 $\times$
Zhaowei (CN, micro-drives)	2.6	10 $\times$
Regal Rexnord / Kollmorgen (US)	10.7	1.8 $\times$
Schaeffler (DE)	≈ 6.9	0.25 $\times$
Unitree (CN, listed Aug 2026)	≈ 27	88 $\times$
UBTech (HK)	4.8	12 $\times$

Table 5. Component makers' valuations (stockanalysis.com, 5–6 Oct 2026). Mature makers trade at 0.3–3.5 $\times$ sales; "humanoid" names at 4.5–10 $\times$; pure plays at 75–88 $\times$ look like a bubble (Unitree opened +629 % and has since fallen ≈ 59 %).

EV / sales multiple	Revenue needed for US\$10 bn	Actuators / yr at US\$250	Share of 2030 market	Share of 2035 market (high case)
2x	US\$5.0 bn	20 M	48 %	5.7 %
4x	US\$2.5 bn	10 M	24 %	2.9 %
10x	US\$1.0 bn	4 M	10 %	1.1 %

Table 6. What a US\$10 bn actuator company needs (ESTIMATED).

Private humanoid makers are valued far higher (Figure $\approx$ US\$39 bn, Apptronik US\$5.5 bn) on expectations, not revenue. A component supplier is valued on revenue and margin. The path to a large outcome is therefore **volume + share in a market that is $\approx$ 10x bigger in 2035 than in 2030**, which means being one of the few qualified non-Chinese suppliers when Western and Indian OEMs reach volume.

9. Go-to-market strategy

9.1 Positioning

"India's actuator company: RobStride-class joints, made and supported in India, with a supply chain that does not depend on Chinese finished goods." You cannot win on price against ¥1,000 Chinese joints. You can win on what Chinese suppliers cannot offer some buyers:

- **Trust and compliance:** defence, government labs and Western buyers who must avoid Chinese robots or components.
- **Local support:** stock in India, a 2-day replacement, Indian invoices with GST credit, engineers who answer the phone.
- **Customisation:** custom ratios, shafts, mounting, IP rating, 72 V windings, bigger sizes (the 360 N·m class nobody sells cheaply).
- **Software:** RobStride-compatible CAN protocol (drop-in), a ready ROS 2 driver, accurate MuJoCo/Isaac actuator models for RL.

9.2 Beachhead customers (in order)

1. **Indian university and research labs** (IITs, IISc, IIITs, ARTPARK): they pay ₹34k–1.05 lakh per imported joint today and need GST invoices and fast service.
2. **Indian robotics startups** (quadrupeds, humanoids, cobots, exoskeletons: Perceptyne, General Autonomy, Svaya, Bhairav, Addverb's suppliers): every one of them imports joints.
3. **Defence** (iDEX challenges, DRDO labs, Zen-type integrators): UGVs, exoskeletons, gimbals, robotic arms: no-China supply is worth a premium.
4. **Export:** Western startups that want a non-China second source.
5. **Industrial servo replacement** ($\approx$ 55k joints/yr imported in India): later, with certifications.

9.3 Product and revenue sequence

Months	What you sell	Why	Milestone
0–6	Integration + resale: RobStride/Damiao joints with an Indian-built CAN interface board, ROS 2 driver, sim models, and an India stock + warranty	Earns cash and customer relationships immediately while your own actuator is developed	10 paying customers; JXA-120 v1 on the dyno
6–18	JXA-120 and JXA-40 , hand-assembled (architecture A), sold to labs and startups; first iDEX application	Learn quality on low volumes with forgiving customers	600 units/yr run-rate; 200 h life data; seed round
18–36	JXA-360 + a 72 V line; die-cast housings, own driver PCB; first export customers	The big joint has no cheap competitor; export lifts price	6,000 units/yr; ISO 9001; Series A
36+	Volume contracts with humanoid OEMs; automated winding and EOL test; second source for magnets (Indian REPM plants)	Scale only against signed demand	60,000 units/yr; Series B or strategic partner

9.4 Metrics to watch and kill criteria

- Gross margin per unit $\geq 35\%$ after Stage 1; field failure rate $< 1\%$ /yr; repeat orders from $\geq 50\%$ of customers.
- Kill or pivot criteria: if after 18 months fewer than 5 customers re-order, or if JXA-120 cannot reach $\leq ₹20k$ cost at 600/yr, switch to being an integrator/distributor + software company (still a business) instead of a manufacturer.

10. Risks

Risk	Mitigation
Chinese price cuts (RobStride guides 300k units in 2026)	Never compete on price alone; sell compliance, service, custom and large joints
Rare-earth magnet export licences (wider Chinese rules paused only until 10 Nov 2026)	Stockpile magnet sets; qualify N48H (no Dy); watch India's REPM plants (2027-28)
Humanoid hype cycle (Unitree -59 % from its IPO open)	Sell to non-humanoid robots too; keep burn low; raise on revenue
Quality / field failures	End-of-line testing on every unit, life testing, warranty reserves
Thin Indian VC for hardware (US\$53 M for all robotics in 2025)	Use grants, defence programmes, customer prepayments, strategic investors

11. Recommendation

1. **Yes, build an actuator company, not another humanoid company.** The demand is real and every Indian robot maker imports today.
2. **Start as an integrator** (sell imported joints with Indian software, stock and support) to earn revenue in months, while JXA-120/40 are developed with the ₹3-4 lakh garage kit from the technical paper.
3. **Sell first to labs, startups and defence**, at prices between RobStride-landed and CubeMars-India; use grants (Seed Fund, incubators, iDEX) before VC.
4. **Raise in small steps** ($\approx ₹2$ cr seed, $\approx ₹11$ cr Series A) and build volume tooling only against signed orders.
5. **Forget "1 % of a trillion"**: aim to be one of the qualified non-Chinese actuator suppliers by 2030. If the 2035 market reaches US\$40-90 bn, that position is worth far more than any shortcut.
6. **Copper is not your lever; magnets and quality are.**

Sources

- `research/raw/market_outlook_2030_copper_raw.md`: humanoid forecasts (BofA Institute "Physical AI" Mar 2026, Citi GPS, Goldman, Morgan Stanley, UBS), IFR World Robotics 2026, S&P Global copper study (Jan 2026), ICSG, COMEX, Stanford AI Index 2026, BLS 2024-34 projections, Brynjolfsson et al., WEF Future of Jobs 2025, valuations (stockanalysis.com)
- `research/raw/india_robotics_industry_capital_raw.md`: Indian actuator/robot companies, clusters, motor-line capex, government schemes (PIB: RDI scheme, ECMS), Indian VC data (Tracxn via press)
- `research/raw/china_actuator_manufacturing_economics_raw.md`: RobStride, Unitree prospectus, broker teardowns, wages, magnet export rules, India REPM scheme, duties
- `actuators/inhouse/`: the technical paper, design and cost models used for unit costs
- `business/actuator_business_model.py`: the staged capital model (all assumptions in the code)